Beal's conjecture division by C

Problem

If , where A, B, C, x, y and z are positive integers and x, y and z are all greater than 2, then A, B and C must have a common prime factor

Abstract

This paper, tries, to prove, that Beal's conjecture, doesn't hold, because if we divide A^x+B^y , by C, should return integer, and be on form $C*C*C*C...$ (power of C). Since this doesn't hold, $A^x + B^y$ can't be on form of powers of C.

Begin of proof

Assumptions

$$P := \{p \in \mathbb{N} : p \text{ is prime}\}$$
$$C := \{c \in \mathbb{N} : c \text{ is composite}\}$$
$$\mathbb{Z}_{>0} := \{n \in \mathbb{Z} : n > 0\}$$
$$\mathbb{Z}_{>2} := \{n \in \mathbb{Z} : n > 2\}.$$

$$p \in P$$
$$c_1, c_2, c_3 \in C$$
$$A, B, C \in \mathbb{Z}_{>0}$$
$$x, y, z \in \mathbb{Z}_{>2}.$$
$$w \in \mathbb{Z}_{>0}.$$

$$A^x = (p * c_1)^x$$
$$B^y = (p * c_2)^y$$
$$C^z = (p * c_3)^z$$

Algebra

$$A^x + B^y = C^z \tag{1}$$

divide both sides by C

$$\frac{A^x}{C} + \frac{B^y}{C} = C^{z-1} \tag{2}$$

From equation (1), we obtain

$$(p * c_1)^x + (p * c_2)^y = (p * c_3)^z \tag{3}$$

convert A,B,C to (3) form.Divide C (which is $p * c_3$), on both sides. Use C form on right side

$$\frac{p^x * c_1^x}{p * c_3} + \frac{p^y * c_2^y}{p * c_3} = C^{z-1} \tag{4}$$

develop left side

$$p^{x-1} * \frac{c_1^x}{c_3} + p^{y-1} * \frac{c_2^y}{c_3} = C^{z-1} \tag{5}$$

multiply both sides by c_3

$$p^{x-1} * c_1^x + p^{y-1} * c_2^y = C^{z-1} * c_3 \tag{6}$$

multiply both sides by C

$$C \underbrace{(p^{x-1} * c_1^x + p^{y-1} * c_2^y)}_{\text{Smaller than } C^z} = C^z * c_3 \tag{7}$$

Since on (7) parentsys are smaller than C^z they are in the form $C^z - w$
And is integer

$$C(C^z - w) = C^z * c_3 \tag{8}$$

divide both sides by C^z.

$$\frac{C}{C^z}(C^z - w) = c_3 \tag{9}$$

develop left side

$$C - \frac{w * C}{C^z} = c_3 \tag{10}$$

put C in evidence

$$C(1 - \frac{w}{C^z}) = c_3 \tag{11}$$

divide both sides by $(1 - \frac{w}{C^z})$

$$C = \frac{c_3}{(1 - \frac{w}{C^z})} \tag{12}$$

convert C left side

$$p * c_3 = \frac{c_3}{(1 - \frac{w}{C^z})} \tag{13}$$

doesn't hold, left side > right side

Conclusion

Since C, doesnt divide, $A^x + B^y$, $A^x + B^y$, can't be on C^z form. So conjecture doesn't hold.

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